

Flood Risk Prediction In Chikwawa District, Malawi:

A machine learning solution

A PROJECT SUBMITTED TO THE FACULTY OF SCIENCE TECHNOLOGY
AND INNOVATION DEPARTMENT OF INFORMATION AND
COMMUNICATION TECHNOLOGY

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Date of submission: 5th May, 2026.

Abstract

Chikwawa District in Malawi's Lower Shire Valley is one of the most flood-prone regions in sub-Saharan Africa. The district faces recurrent and devastating flooding that severely impacts agriculture, health, and local livelihoods. Existing early warning systems often lack spatial resolution and lead time necessary for effective community-level action. This project proposes a localized machine learning solution to predict flood risk using multi-source environmental and historical data. The system integrates data including historical rainfall patterns, Shire River discharge levels, soil moisture estimates, and satellite-derived topographic indices (NDVI, TWI) sourced from Google Earth Engine (GEE), the Department of Climate Change and Meteorological Services (DCCMS), and the Department of Disaster Management Affairs (DoDMA).

By leveraging advanced classification algorithms such as Random Forest, XGBoost, and LightGBM, the system generates high-resolution, probabilistic flood risk forecasts with increased accuracy. The final solution provides an interactive dashboard and alert mechanism designed to support disaster management officials in proactive decision-making and resource allocation. This project follows the CRISP-DM methodology to ensure a robust, iterative development process that addresses the specific data scarcity and computational constraints of rural Malawi.

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Introduction

1.0 Background

Chikwawa District, located in Malawi's southern region within the Lower Shire Valley, is one of the most flood-vulnerable areas in sub-Saharan Africa[1]. The district's exposure to flooding results from its unique topography, where intense rainfall in the surrounding highlands generates rapid runoff that overwhelms the low-lying plains. This geographical setting, combined with socio-economic challenges and the growing influence of climate change, has made flooding a recurrent and increasingly severe threat.

In recent years, the frequency and intensity of extreme weather events have risen sharply. Successive cyclones; Idai in 2019, Ana in 2022, and Freddy in 2023[2], struck within a short period, causing devastating flash floods and prolonged floods across the district.

The consequences of these floods are far-reaching. Agriculture, the backbone of local livelihoods, suffers heavy losses, with smallholder farmers often losing between 70% and 100% of their maize, sorghum, and rice harvests[3]. This regularly triggers serious food insecurity. Furthermore, floodwaters left standing for extended periods become breeding sites for disease vectors and significantly increase the spread of waterborne illnesses. The 2019 floods, for example, led to over 1,200 cholera cases and claimed 38 lives in Chikwawa within just one week[4].

At present, flood forecasting in Malawi is primarily handled by the Department of Climate Change and Meteorological Services (DCCMS) through Numerical Weather Prediction (NWP) models[5]. While useful, these traditional approaches face constraints in terms of timeliness, precision, and ability to handle complex local conditions effectively.

Machine Learning presents a promising solution to overcome these limitations. By learning patterns from historical meteorological, hydrological, and environmental data, ML models can generate faster, more accurate, and localized flood risk predictions. Such systems have the potential to strengthen early warning mechanisms, support better land management decisions, improve evacuation planning, and enhance overall community resilience in flood-prone regions like Chikwawa.

1.1 Problem Statement

Chikwawa District remains one of the most severely affected flood-prone areas in Malawi, yet communities continue to experience frequent and devastating flooding with insufficient advance warning. Existing forecasting systems often fail to deliver timely, accurate, and locally relevant information, leaving residents with little time to prepare or respond effectively.

The Lower Shire Valley's distinctive geography — characterized by extremely low-lying plains and the massive Shire River, the only outlet of Lake Malawi — creates a highly complex and localized flooding pattern[6]. Heavy runoff from surrounding highlands, combined with factors such as rapid river level rises, backflow from tributaries, localized intense rainfall, river siltation, deforestation, and expanding settlements, produces flood events that national-level models struggle to predict with the necessary precision and granularity.

Current early warning mechanisms in Malawi are largely centralized and depend on broad meteorological and hydrological models[7]. These generalized approaches do not adequately account for Chikwawa-specific triggers, resulting in warnings that frequently arrive too late, lack village-level detail, and fail to reach remote rural communities in time for meaningful action.

The impacts of this predictive shortfall are both severe and recurring:

- Delayed or ineffective evacuations, leading to preventable loss of human lives
- Extensive destruction of homes, schools, health facilities, roads, and bridges
- Massive losses of staple crops (maize, sorghum, cotton, sugarcane) and livestock, pushing farming households into food insecurity and debt
- Large-scale displacement of populations into temporary camps
- Increased outbreaks of waterborne diseases such as cholera and malaria due to contaminated water
- Long-term economic setbacks as families lose productive assets and struggle to recover season after season

There is therefore a pressing need for a localized, machine learning-driven flood risk prediction system specifically designed for Chikwawa District. By integrating historical and real-time data — including rainfall intensity, Shire River discharge, soil moisture levels, land-use changes, elevation, and past flood records — such a system can generate accurate, hyper-localized, and timely warnings several days in advance for specific communities, extension workers, and disaster management officials.

1.2 Scope of the Development Project

This project involves the development of a Machine Learning-based Flood Risk Prediction System for Chikwawa District. The system will utilize data science and machine learning techniques to provide timely and localized flood risk predictions to support early warning and decision-making.

1.2.1 Objectives

- To develop an accurate flood risk prediction model using classification techniques.
- To build a user-friendly dashboard for visualization and decision support.
- To establish a data processing pipeline that handles meteorological and hydrological data effectively.
- To enable role-based access for relevant stakeholders, particularly DCCMS administrators.

1.2.2 Features Included

The following functionalities will be implemented:

- Data Preprocessing Pipeline; Conversion of TIFF files and other raw datasets into structured CSV format, data cleaning, and feature engineering.
- Predictive Analytics: Development of classification models for flood risk prediction using:
 - Random Forest Classifier
 - XGBoost Classifier
 - LightGBM Classifier
- Model Training and Evaluation: Training selected models and comprehensive performance evaluation using Accuracy, Precision, Recall, and F1-Score.

- Visualization and Dashboards: Interactive dashboards with visual analytics, including threshold-based flood risk visualization and historical findings.
- Role-Based Access Control: Secure login system with different access levels (Admin access for DCCMS staff).
- Real-time Data Integration: Integration with existing systems and APIs for ingesting real-time or near real-time data.
- Reporting: Generation of model performance reports and flood risk insights.

1.2.3 Features Excluded

The following items are not part of the current project scope:

- Development of a mobile application with offline functionality.
- Implementation of Deep Learning models (e.g., CNN, LSTM, etc.).
- Hardware procurement or installation of new sensors.
- Nationwide deployment (limited to Chikwawa District in this phase).
- Automated physical response mechanisms (e.g., sirens or dam control).

1.2.4 Operational Environment

The system will be designed as a web-based application that can integrate with existing tools and platforms used by the Department of Climate Change and Meteorological Services (DCCMS) through API integration. It will operate in a secure, role-based environment to ensure data confidentiality and controlled access.

1.4 System personnel

Role	Description
System Users	Department of Climate Change and Meteorological Services (DCCMS) officials
Developer(s)	ZAHEERAH ABDULRASHID, RODRICK KAPOPO, FAVOUR KAMPANGIRE, LOUIS MWAMSAKO, JONATHAN BANDA, GEORGE MAPELERA, LINGALITHU KADEWELE

System Owner	Department of ICT, Mzuzu University
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System Requirements

2.0 System Overview

The current flood monitoring system in Chikwawa District, Malawi is largely manual and reactive. Authorities and residents rely on historical experience, delayed weather reports, and physical observation of river levels. This approach often results in delayed warnings and inadequate preparedness, increasing the risk of property damage and loss of life.

The proposed system is a real time flood risk prediction system that applies data science and machine learning techniques to analyze historical and environmental data. The system will process datasets such as rainfall and river levels, perform preprocessing and Exploration Data Analysis (EDA), and generate predictive insights. The results will be presented through a dashboard and alert system to support early decision making and disaster preparedness.

2.1 Functional Requirements

ID	Function Name	Description
FR1	User Authentication	The system shall allow users, especially administrators, to log in securely.
FR2	Data Upload	The system shall allow uploading of datasets such as CSV files.
FR3	Data Preprocessing	The system shall clean and transform raw data including handling missing values and converting TIFF files to CSV.
FR4	Role Based Access	The system shall restrict certain features to administrators only.
FR5	Prediction Engine	The system shall accept input data including API data and generate flood risk predictions.

FR6	Error Handling	The system shall detect and handle errors such as incorrect login credentials and invalid data.
FR7	Alert System	The system shall send alerts such as SMS notifications to officials and residents when high flood risk is detected.

2.2 Interface Requirements

- **Data Interface:** The system shall support importing datasets in CSV format. It shall allow integration with external data sources such as APIs for weather and environmental data.
- **User Interface:** The system shall provide a dashboard for visualization of flood risk levels. It shall display charts, graphs, and alerts in a clear and simple format. The interface should be user friendly for both technical and non-technical users.
- **External Interfaces:** The system shall integrate with external APIs for real time or near real time data input. It should support communication services such as SMS gateways for sending alerts.
- **Storage Interface:** The system shall store both raw and processed data. It shall maintain prediction results for future reference and analysis.

2.3 Non-Functional Requirements

2.3.1 System-Related Non-Functional Requirements

ID	Function Name	Description
NFR1	Performance	The system shall generate flood risk predictions from preprocessed input data in less than 10 seconds to ensure timely alerts.
NFR2	Reliability	The prediction engine and dashboard shall maintain an uptime of at least 99.5% during the peak flood season (November to April).

NFR3	Scalability	The data pipeline shall be capable of processing an increase in historical records (up to 50,000 records) without a significant increase in processing latency.
NFR4	Security	Access to the administrative dashboard and sensitive raw datasets shall be restricted via secure login and role-based access control (RBAC).

2.3.2 User-Related Non-Functional Requirements

ID	Function Name	Description
UR1	Usability	The dashboard interface shall be intuitive, allowing DCCMS officials to interpret risk heatmaps and charts without requiring extensive machine learning expertise.
UR2	Supportability	The system shall include comprehensive technical documentation and a user manual to facilitate future maintenance by district ICT personnel.
UR3	Availability	The prediction results and interactive visualizations shall be accessible via standard web browsers

		(Chrome, Firefox, Edge) on both desktop and mobile devices.
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2.4 System Interfaces

Interface Type	Description
API Interfaces	GET /weather-data, POST /predict, GET /alerts. Data format: JSON.
CLI Interfaces	Commands: python train_model.py, python predict.py --input data.csv
Dashboard UI	Features: Risk display, maps, graphs, alerts, admin controls.
Hardware Interfaces	Inputs: Sensors (rainfall, river level), weather APIs. Outputs: SMS alerts, dashboard display.

Data Requirements

3.0 Data Collection/Ingestion

Data will be collected from multiple diverse sources to ensure comprehensive coverage:

- Google Earth Engine (GEE) APIs: Used for earth observation datasets (satellite imagery, digital elevation models).
- Department of Climate Change and Meteorological Services (DCCMS): Source for local weather station data and ground-truth rainfall.
- Department of Disaster Management Affairs (DoDMA): Source for historical flood impact reports and labels.

Frequency: Meteorological data will be collected at daily intervals, while satellite imagery (e.g., Sentinel radar) will be ingested based on satellite overpass schedules (e.g., every 6-12 days).

Volume: Raw spatial datasets processed on GEE involve multi-gigabyte files, but the final exported tabular datasets will be lightweight (approximately 100MB - 500MB).

Access Protocols: Data will be ingested using the GEE Python API (ee and geemap libraries) using OAuth 2.0 authentication. Local meteorological data will be processed via CSV/Excel exports.

3.1 Data Description

- Source: GEE (NASA GPM, Sentinel-1, SRTM DEM), DCCMS, DoDMA.
- Format: GeoJSON for administrative boundaries, GeoTIFF and CSV for the final structured tabular datasets used in modeling.

Variables/fields and their types:

- Latitude (float).
- Longitude (float).
- Elevation_m (float): Digital Elevation Model values.
- Slope_deg (float).
- NDVI: Normalized Difference Vegetation Index (float).
- LandCover (integer).
- Rainfall_mm (float).
- Dist_River_m (float).
- Flood (integer): Target variable (1 for flood, 0 for no flood).
- Event (object).

Sample size and limitations: The dataset covers 3-11 years of historical data, yielding approximately 18,000 daily records in all observed zones. A significant limitation is the severe class imbalance (few actual flood events compared to non-flood days) and potential gaps in ground-station records requiring imputation from satellite estimates.

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the first major step of our machine learning work. The main goal of this phase is to find hidden patterns in past flood events, rainfall records and elevation data. By looking closely at this data, we want to understand how water flows and causes floods in the Lower Shire Valley.

If we explore the data carefully, we will successfully find important trends over time and see which areas in Chikwawa are most at risk. We will mostly use data from major storms, like Cyclone Idai in 2019 and Cyclone Freddy in 2023. We will use visual tools and create charts, such as histograms, box plots and scatter plots. These

charts will help us compare rainfall amounts with the Shire River water levels. Also, we must find extreme weather events and data errors early. Therefore, we will use outlier detection methods so that bad records do not confuse our machine learning models.

Our key EDA activities will include the following steps:

- Temporal trend analysis - we will look at past rainfall and river level data over time to find seasonal patterns and peak flood seasons.
- Spatial data visualization - we will use GIS tools to map past floods across low-lying areas to see which Traditional Authorities are in the most danger.
- Anomaly and outlier detection - we will find extreme weather patterns and possible sensor errors that could make our data incorrect.
- Correlation assessment - we will use statistical tools, like Pearson or Spearman coefficients, to see how heavy rain in the higher lands connects to flash floods in the lower lands.

3.3 Data Cleaning and Preprocessing

Data cleaning and preprocessing are very important steps to prepare good, high-quality data for our machine learning models. Because our project uses data from both satellites and ground sensors, the raw data will naturally have some noise and matching issues.

Raw weather data often has missing value or errors, and we must fix these problems quickly to keep our predictions accurate. If weather stations fail to record daily data because of power cuts or damage, we will use techniques like K-Nearest Neighbors (KNN) imputation or time-based interpolation to fill in the missing gaps. Also, the data for the Shire River water levels will be matched perfectly in time with the rainfall records. Since different data types use different units (for example, elevation in meters and river flow in cubic meters per second), we will use standard scaling tools to normalize all the numbers.

The data preprocessing phase includes the following steps in order:

- Data cleaning and imputation - we will handle missing values using interpolation, remove duplicate records and fix impossible numbers like negative rainfall.

- Spatial resampling and alignment - we will make sure all map data has the same resolution, keeping it at a strict 1 km scale.
- Feature engineering - we will create new variables that help predictions, like calculating total rainfall over the past 3, 7 and 14 days and estimating soil moisture.
- Data normalization - we will scale all number values to a standard range using Min-Max or Standard Scalers so that larger numbers do not overpower the machine learning model.

3.4 Data Summary

- Dataset size: The final combined dataset ready for modeling is expected to be under 100 MB.
- Types and dimensions: Tabular (2D array/DataFrame).
- Data dimensions: 14,000+ rows (in total 18,000 rows) by 9 columns (features).
- Key variables: Elavation_m, Slope_deg, NDVI, Rainfall_mm, Dist_River_m, topographic_wet_index, elevation_to_river, PS_codes (after scaling Permeability_scores), LC_codes (encoded LandCover scores) and Flood.

Technical Requirements

4.0 Software and Tools

The development of the Chikwawa flood prediction system relies entirely on robust, open-source programming languages, libraries, and frameworks. This ensures sustainability, reproducibility, and cost-effectiveness.

Programming Language:

- Python 3.x: The primary programming language used for the entire pipeline, from data extraction to machine learning model development.

Machine Learning & Data Processing Libraries:

- Scikit-learn: Utilized for baseline model development (Random Forest), data splitting (train/test sets), and evaluation metrics (accuracy, precision, recall, F1-score, confusion matrix).

- XGBoost & LightGBM: Advanced gradient boosting frameworks used for the core predictive modeling, chosen for their high performance on tabular data and ability to handle nonlinear relationships.
- Imbalanced-learn: Used to apply techniques like SMOTE to handle the imbalanced nature of flood data[8] (i.e., fewer flood events compared to non-flood days).
- Pandas: Essential for data manipulation, cleaning, temporal alignment, and structuring the final dataset into a tabular format for modeling.

Geospatial & Earth Observation Libraries:

- GeoPandas & Shapely: Used for handling geospatial geometries, defining district boundaries (GeoJSON), and spatial masking.
- EarthEngine-API (Google Earth Engine) & Geemap: Used to interface with Google's cloud infrastructure to access and process massive petabyte-scale environmental datasets (GPM rainfall, Sentinel radar, topography) without downloading the raw data locally.
- Rasterio & Xarray: Used for handling and resampling gridded raster datasets (like Digital Elevation Models).
- OSMnx: Utilized for extracting localized infrastructure and spatial networks.

Development Environment & Visualization:

- Jupyter Notebooks: The primary interactive development environment, enabling step-by-step data processing, experimentation, and documentation.
- Matplotlib & Seaborn: Used for exploratory data analysis (EDA), visualizing feature importance, and plotting evaluation matrices.

4.2 Hardware Requirements

The system utilizes a hybrid approach, distributing computational load between local hardware and cloud platforms.

Local Computing Requirements (Data assembly, Feature Engineering, and Model Training):

- RAM: Minimum 16 GB (32 GB recommended). Geospatial processing requires substantial memory.
- Storage: 50 GB to 100 GB of free space, preferably on an SSD to ensure fast read/write speeds for csv and geojson data assets.
- CPU: Multi-core processor (e.g., Intel i5/i7 or AMD Ryzen equivalent) to handle parallel processing during model hyperparameter tuning.

Cloud Computing Environment:

- Google Earth Engine (GEE): Used to offload the heaviest computational tasks (like temporal aggregation of satellite imagery). GEE handles the processing on Google's cloud servers, meaning local hardware only needs to download the aggregated, lightweight tabular results. A stable internet connection is required to communicate with the GEE servers.

Ethics and Privacy

Because this project aims to help people, we are strongly committed to following strict ethical rules while handling environmental and community data. The main goal of this flood prediction model is to save lives and protect properties; therefore, doing things the right way is very important.

Our proposed early warning system mostly uses weather and map data, which does not easily threaten personal privacy. However, if we include local community data in the future, it will be protected very carefully. If we conduct surveys about how disasters affect specific households, we will immediately remove all personal details like names and addresses. We must also make sure that our machine learning models are trained fairly across all areas of Chikwawa so that no remote village is unfairly ignored by the computer predictions. In the end, all our data work will be open, fair and responsible.

To guide our research, our ethical framework will follow these important rules:

- Data ownership and usage - all weather and water data belong to the respective government agencies, and we will only use it for school research and helping the community.
- Privacy protection and anonymization - if we collect any survey data from communities during our testing, we will apply strict rules to make sure everyone stays anonymous.

- Algorithmic fairness and bias checking - we will actively watch for and fix any bias in the machine learning model. This ensures our flood risk checks are fair and highly accurate across all the different Traditional Authorities in the district.
- Regulatory compliance - all our work and computer processes will always follow the ethical guidelines of Mzuzu University and the national data policies.

Project Plan

6.0 Estimated Development Costs

Category	Description	Estimated Cost (USD)
Hardware	High-performance laptops (16-32GB RAM) for local processing	\$2,500 equivalent to MK 4,353,750.00
Software	Python, QGIS, Scikit-learn, GEE	\$0 (Open Source)
Data Acquisition	Field visits for ground-truthing, transport to Chikwawa	\$344 equivalent to MK599,076.00
Cloud / Hosting	Future cloud deployment (AWS/Azure) for a live dashboard	\$300 (Annual) equivalent to MK522,450.00
Personnel	Stipends/allowances for data science student researchers	\$200 equivalent to MK348,300.00
Contingency	Unforeseen expenses (internet data bundles, API limits)	\$100 equivalent to MK174,150.00
Total Estimated Cost		\$3,444 = MK 5,997,726.00

6.1 System Development Life Cycle (SDLC) Approach

The project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, an iterative approach tailored for machine learning and data science projects:

- Business/Domain Understanding: Defining the flood challenges specific to the Chikwawa district and setting predictive objectives.
- Data Understanding: Sourcing data via GEE, local meteorological records, and historical flood reports.
- Data Preparation: The most intensive phase, involving spatial resampling, handling missing values, scaling, and engineering lagging rainfall features.

- Modeling: Training baseline (Random Forest) and advanced (XGBoost, LightGBM) classifiers.
- Evaluation: Assessing models against validation sets using precision, recall, and F1-scores.
- Deployment & Recommendations: Formulating a prototype workflow that disaster management teams can use to generate early warnings.

6.2 Systems Delivery Specifications

The final deliverables for this project will include:

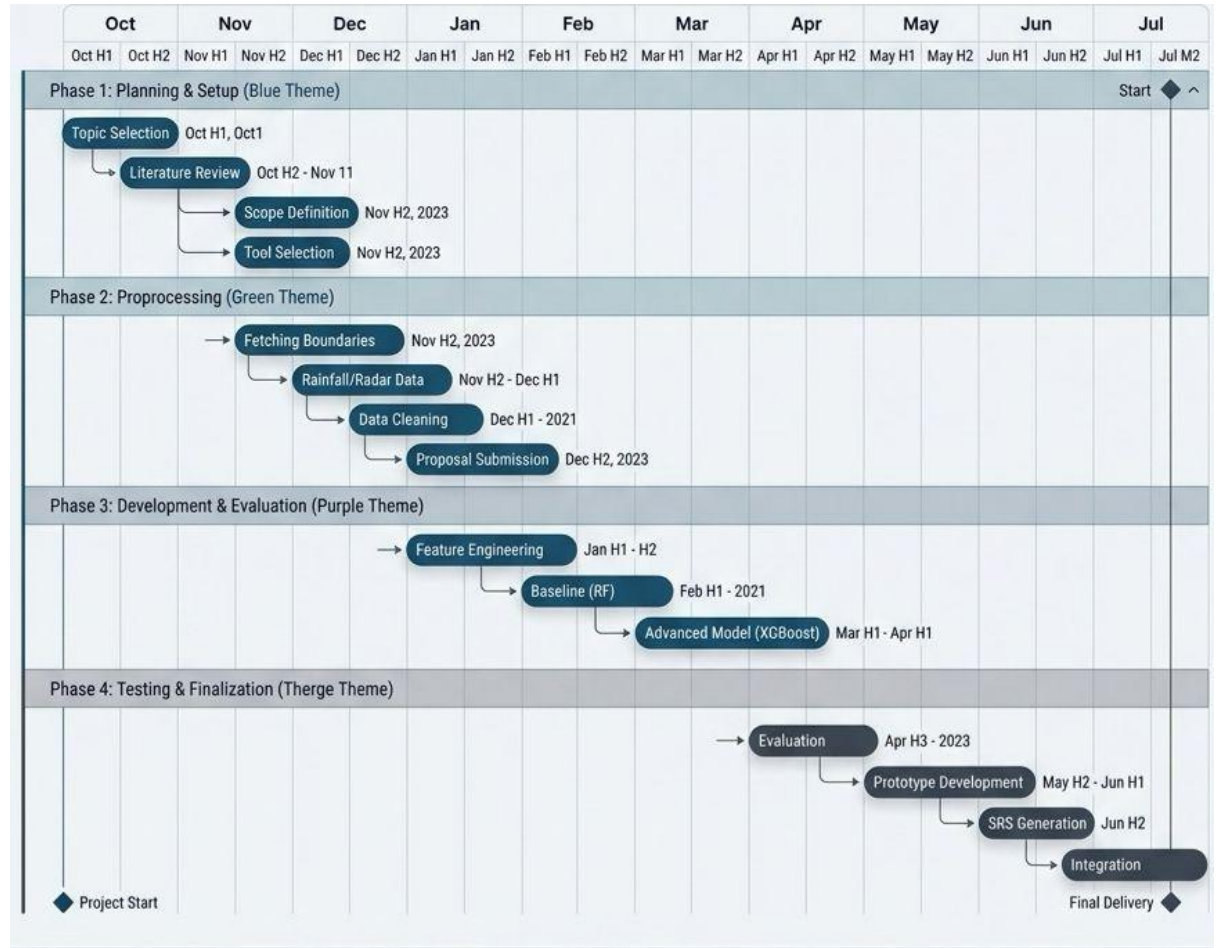
- Processed Datasets: Cleaned, combined tabular data (chikwawa_flood_dataset.csv) ready for modeling.
- Codebase / Notebooks: A fully documented sequence of Jupyter Notebooks mapping the entire pipeline from boundary fetching to feature extraction.
- Trained Predictive Models: Exported LightGBM/XGBoost models capable of ingesting new meteorological data.
- Final Research Report: A comprehensive document detailing methodology, model evaluation, and recommendations for integration into Chikwawa's early warning framework.

6.3 Installation and Access Requirements

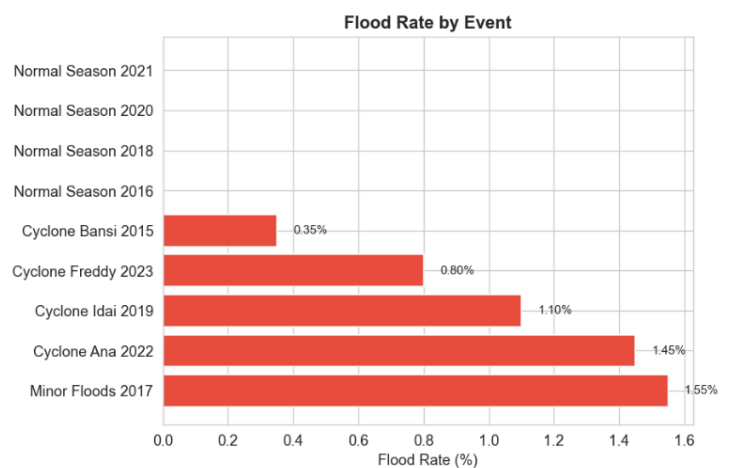
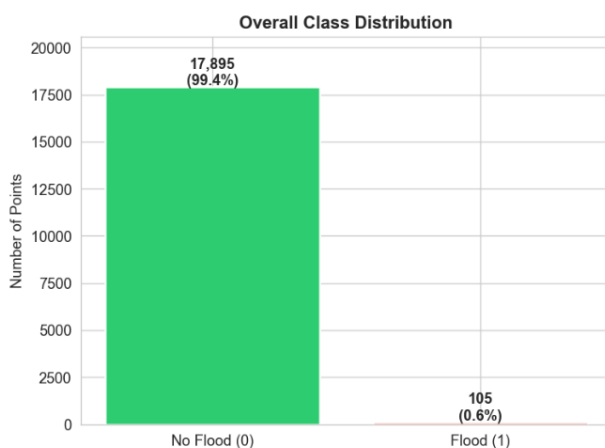
- Software Dependencies: The system requires Python 3.8 or higher. Key libraries include earthengine-api, geemap, pandas, scikit-learn, xgboost, and lightgbm.
- Environment Setup:
 - Clone the project repository from the official GitHub source.
 - Install all required Python packages using the command: `pip install -r requirements.txt`.
 - Initialize and authenticate the Google Earth Engine API using: `earthengine authenticate`.
- System Access: The final application will be deployed on a secure web server. Authorized users will access the system via a provided URL and their unique credentials. Developer access to the codebase will be managed through a private repository.

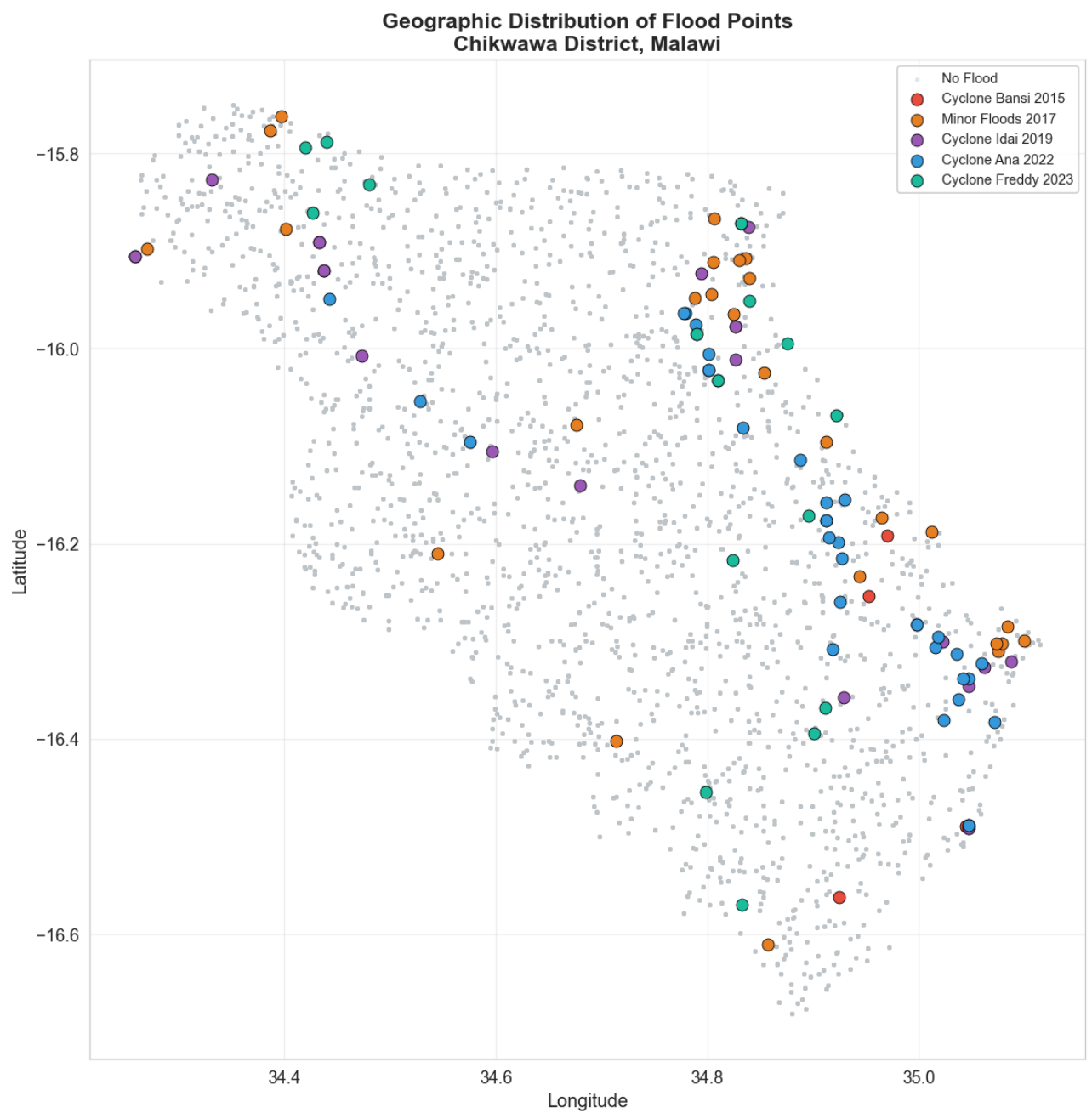
6.4 Project Timeline (Gantt Chart)

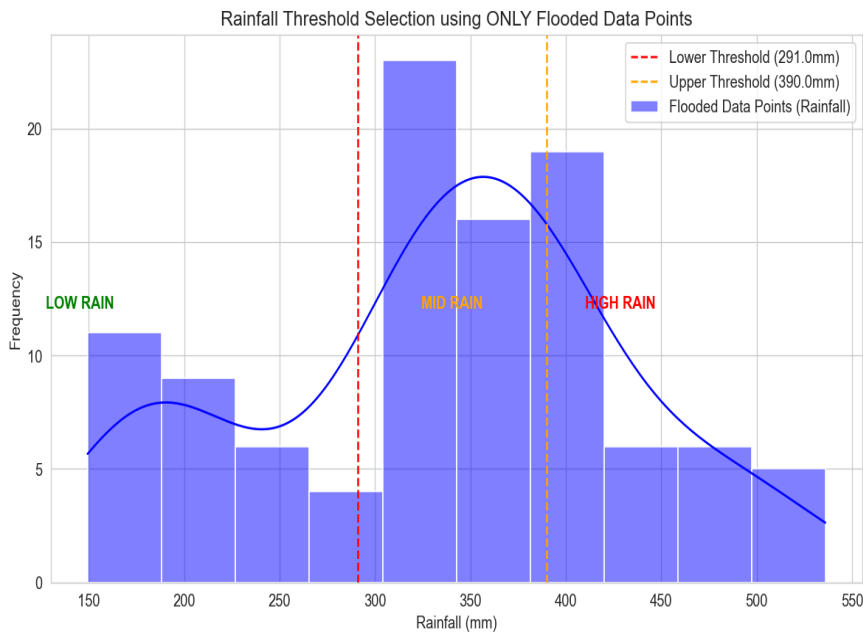
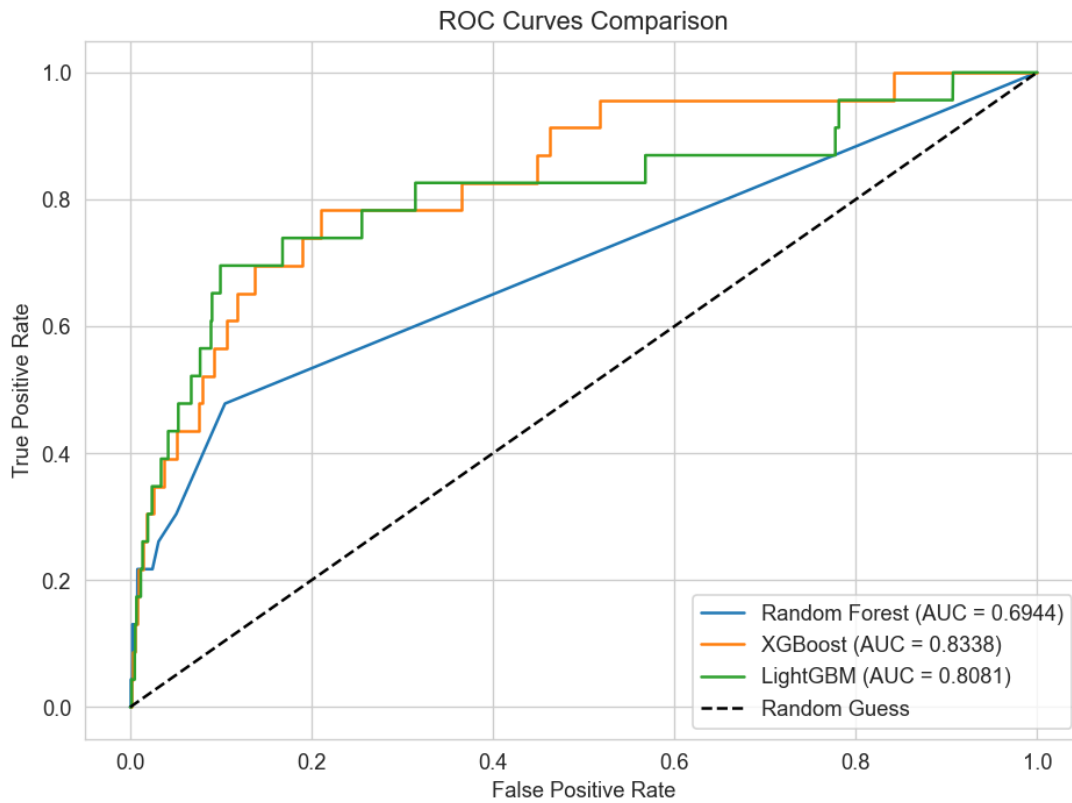
The following Gantt chart outlines the chronological phases of the project across the 9-month development cycle. **H1** represents First Half of a month (1st-15th of the month). **H2** represents Second Half of a month (16th through the end of the month).



Appendices







Confusion Matrix table

	Accuracy	Precision	recall	F1_Score	Auc
Random Forest	0.604642	0.147059	0.217391	0.175439	0.694394
LightGBM	0.604083	0.131579	0.217391	0.163934	0.833842
XGBoost	0.623585	0.109091	0.26087	0.153846	0.808128

Glossary

TERM	DEFINITION
DCCM	Department of Climate Change and Meteorological Services (Malawi).
DoDMA	Department of Disaster Management Affairs (Malawi).
GEE	Google Earth Engine; a cloud platform for earth observation data analysis.
NDVI	Normalized Difference Vegetation Index; a measure of vegetation density and health.
NWP	Numerical Weather Prediction; mathematical models used for weather forecasting.
SMOTE	Synthetic Minority Over-sampling Technique; used to balance flood vs. non-flood data.
TWI	Topographic Wetness Index; used to identify areas where water is likely to accumulate.
SRTM	Shuttle Radar Topography Mission; providing digital elevation model (DEM) data.
CRISP_DM	Cross-Industry Standard Process for Data Mining; the standard methodology used for this project.

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